

Keywords: grokking • learning theory • ridge regression • generalization dynamics

To Grok Grokking

The First Mathematical Proof That Delayed Generalization Is an Inevitable Feature of Over-Parameterized Regularized Learning

By Mingyue Xu, Gal Vardi & Itay Safran (Hebrew University of Jerusalem; Ben-Gurion University of the Negev)

arXiv:2601.19791 • ICML 2026 Honorable Mention

Grokking—the strange phenomenon where a neural network first memorizes its training data and then, thousands of steps later, suddenly generalizes —has captivated the machine learning community since its discovery. Yet the phenomenon has resisted rigorous explanation. This paper delivers one: in a classical ridge regression setting, gradient descent with weight decay *provably* must traverse three phases—rapid overfitting, prolonged poor generalization, and eventual near-perfect test accuracy—with explicit quantitative bounds on how long each phase lasts as a function of every hyperparameter.

AT A GLANCE

Problem: Grokking has no rigorous theoretical explanation; practitioners cannot predict when or whether it will occur.

Why it matters: Grokking affects training schedules and regularization choices across many deep learning applications; without a theory, practitioners rely on empirical trial-and-error.

Method: Exact analysis of gradient flow with ℓ_2 regularization on over-parameterized Gaussian linear regression via spectral decomposition of the sample covariance.

Result: End-to-end proof of all three grokking phases; grokking time $T_{\text{grok}} \asymp \lambda^{-1}$ scales inversely with weight decay.

Evidence: Theory-predicted and measured grokking times match within 5% across all tested hyperparameter configurations.

The Big Picture

Grokking was first documented empirically in 2022: small transformers trained on modular arithmetic tasks would rapidly reach 100% training accuracy yet hover near chance-level test accuracy for tens of thousands of additional gradient steps, before abruptly transitioning to near-perfect generalization. The phenomenon was striking because it violated the common understanding that

a model which overfits is simply “stuck”—that continued training without additional data would only worsen generalization.

Subsequent work documented grokking in increasingly diverse settings, from formal grammar to image classification to weight-shared networks. Competing explanations were offered—“efficient circuits” displacing “memorizing circuits,” progressive Fourier feature acquisition, norm minimization dynamics—but all remained heuristic accounts tied to the neural network architecture rather than to the statistical estimation problem underlying them.

This paper strips away the architecture and asks: does grokking occur in the simplest possible supervised learning setting? The answer is yes.

Why Existing Approaches Fall Short

Prior theoretical analyses of grokking were confined to toy settings (e.g., a single neuron or a two-layer network on binary classification) and could not derive exact grokking times. Most empirical studies on grokking used transformers on modular arithmetic, where the discrete structure of the task confounds the statistical learning dynamics.

The “efficient circuit” account holds that grokking reflects the displacement of high-norm memorizing solutions by low-norm generalizing solutions, driven by weight decay. While this story is directionally correct, it does not predict when the displacement occurs or how hyperparameter choices modulate the transition time. The literature on double descent and overfitting-to-generalization transitions in ridge regression provides important background, but none of these works study the *temporal dynamics* of the transition under gradient flow—they analyze stationary solutions only.

Core Mechanism

The mechanism of grokking in ridge regression is the spectral separation between “spurious” and “signal” components of the weight vector.

Early in training, gradient descent minimizes training loss by growing weight components along *all* singular directions of the data matrix, including those with small singular values that correspond to noise. These high-magnitude spurious components allow the model to fit the training data perfectly but generalize poorly.

Weight decay λ then acts as an exponential decay on these components at rate proportional to λ . The signal components (those aligned with large singular values) are also decayed, but they are simultaneously reinforced by the error gradient from training examples that the spurious components cannot fit well. Eventually, as the spurious components shrink below the noise floor, the signal components dominate and generalization improves

rapidly.

Technical Formulation

Let $X \in \mathbb{R}^{n \times d}$ (with $d \gg n$) be the data matrix and $y \in \mathbb{R}^n$ the labels. Gradient flow on the regularized loss $\mathcal{L}(\theta) = \frac{1}{2n} \|X\theta - y\|^2 + \frac{\lambda}{2} \|\theta\|^2$ gives:

$$\dot{\theta}(t) = -\frac{1}{n} X^\top (X\theta(t) - y) - \lambda\theta(t)$$

Decomposing via the SVD $X = U\Sigma V^\top$, with $\bar{y}_i = (U^\top y)_i / \sqrt{n}$ and singular values σ_i , the solution is:

$$\theta(t) = V \operatorname{diag} \left(\frac{\sigma_i^2/n}{\sigma_i^2/n + \lambda} \left(1 - e^{-(\sigma_i^2/n + \lambda)t} \right) \frac{\bar{y}_i}{\sigma_i/\sqrt{n}} \right) \mathbf{1}$$

For small singular values $\sigma_i \rightarrow 0$ (the spurious directions), the weight decays at rate λ once it has reached its early maximum. The grokking time is dominated by this decay:

$$T_{\text{grok}} \asymp \frac{1}{\lambda} \log \frac{\|\theta_{\text{spurious}}^*\|}{\varepsilon}$$

where $\theta_{\text{spurious}}^*$ is the spurious-direction norm at peak overfitting and ε is the target generalization error.

Learning or Inference Procedure

1. Initialize $\theta(0) = 0$.
2. Run gradient flow on $\mathcal{L}(\theta)$ with weight decay $\lambda > 0$.
3. **Phase I (steps 0 to T_{fit}):** Training loss reaches near-zero; test error remains high as spurious components grow.
4. **Phase II (steps T_{fit} to T_{grok}):** Training loss stays near-zero; weight decay erodes spurious components; test error remains high.
5. **Phase III (steps $> T_{\text{grok}}$):** Spurious components drop below noise threshold; signal components dominate; test error drops rapidly to near zero.

The three phases are mathematically sharp: the paper proves that $T_{\text{grok}}/T_{\text{fit}} \rightarrow \infty$ as $\lambda \rightarrow 0$.

What the Guarantee Says

The paper proves the following for over-parameterized ridge regression with $d/n \rightarrow \infty$ and a well-separated signal-to-noise ratio:

- (1) The model *necessarily* overfits: $\mathcal{L}^{\text{train}}(\theta(T_{\text{fit}})) = 0$ with high probability.
- (2) Overfitting *necessarily* persists: there exists $c > 0$ such that $\mathcal{L}^{\text{test}}(\theta(t)) \geq c$ for all $t \in [T_{\text{fit}}, T_{\text{grok}}]$.
- (3) Generalization *eventually* occurs: $\mathcal{L}^{\text{test}}(\theta(t)) \rightarrow 0$ as $t \rightarrow \infty$.

The grokking time $T_{\text{grok}} = \Theta(\lambda^{-1})$ is tight: both the upper and lower bounds are of this order.

Experimental Findings

λ	n	Measured T_{grok}	Predicted
10^{-3}	100	42,800	41,500
10^{-2}	100	4,600	4,300
10^{-1}	100	520	490
10^{-3}	500	39,200	38,000

Theory and experiment agree within 5% across all configurations. The paper also shows that grokking can be *eliminated* entirely by choosing $\lambda \geq \lambda^*$ (derived analytically) or *accelerated* by curriculum learning, where the training order of examples is arranged to minimize the norm of the spurious component.

Ablations and Interpretation

Effect of λ : As λ increases 10-fold, grokking time decreases by approximately 10-fold, confirming the $\Theta(\lambda^{-1})$ prediction. Above a critical $\lambda^* = \sigma_{\min}^2/n$ (smallest signal singular value squared), grokking is eliminated: the model never overfits.

Effect of n : Increasing n (more training data) reduces the norm of the spurious component at overfitting, shortening T_{grok} sublinearly. The theory predicts $T_{\text{grok}} = \Theta(\lambda^{-1} \log n)$ and experiments confirm the logarithmic dependence.

Nonlinear networks: Empirical tests on two-layer ReLU networks trained on the same linear regression task exhibit grokking with qualitatively identical phase structure and λ^{-1} dependence, suggesting the linear analysis captures essential dynamics.

Curriculum learning: Ordering examples by signal-to-noise ratio (easiest first) reduces T_{grok} by 3–4 $\times$ without changing the final test error, consistent with the theory’s prediction that curriculum learning reduces the spurious component norm at overfitting.

Reference

Mingyue Xu, Gal Vardi, Itay Safran. **To Grok Grokking: Provable Grokking in Ridge Regression.** arXiv:2601.19791, January 2026. ICML 2026 Outstanding Paper Honorable Mention. <https://arxiv.org/abs/2601.19791>